

Hogfish

Species: *Lachnolaimus maximus*, Stock code: Hogfish - Florida Keys - East Florida

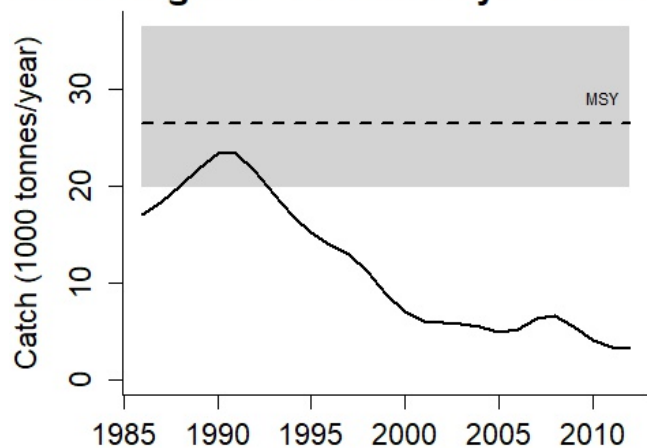
Region: Southeast Shelf

Marine Ecoregion: Florida Keys - East Florida

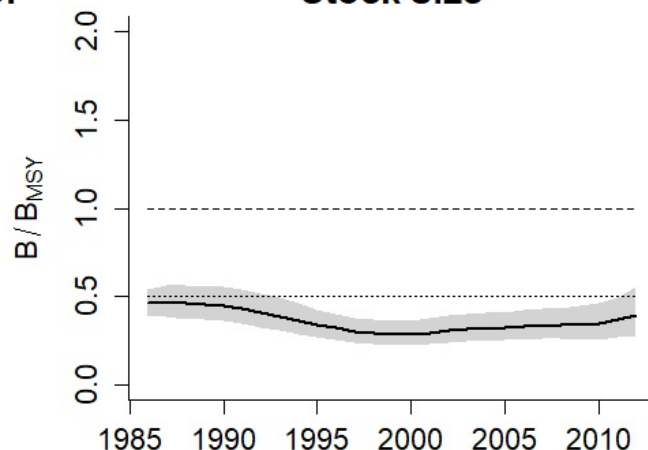
Reconstructed catch data used from years 1986 - 2012

For figure captions and method see <http://www.seaaroundus.org/cmsy-method>

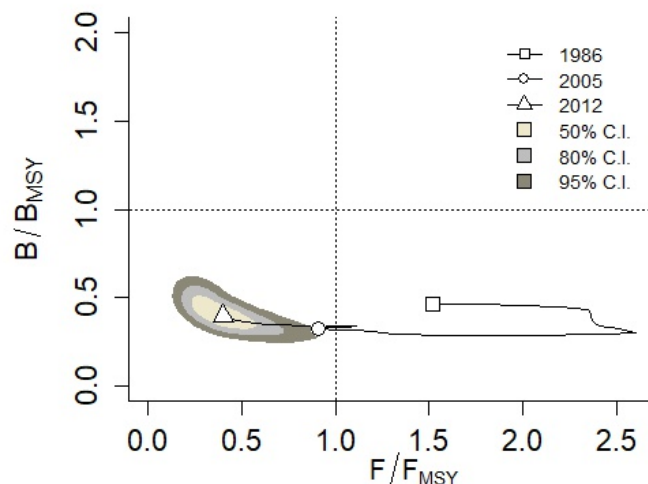
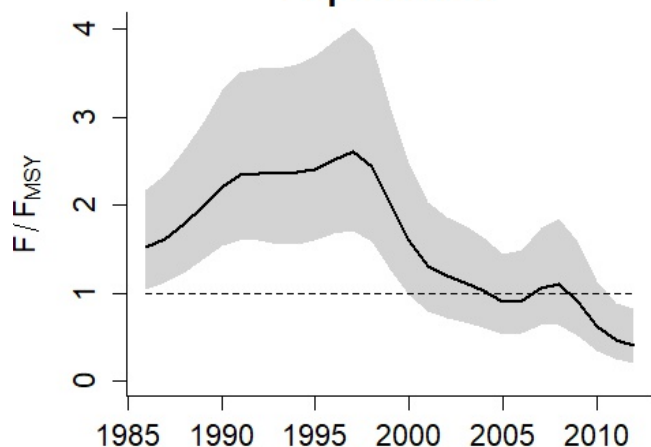
Catch Hogfish - Florida Keys - East Floi



Stock size



Exploitation



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Results for management (based on BSM analysis)

$F_{msy} = 0.0858$, 95% CL = 0.0531 - 0.141 (if $B > 1/2 B_{msy}$ then $F_{msy} = 0.5 r$)

$F_{msy} = 0.0681$, 95% CL = 0.0421 - 0.112 (r and F_{msy} are linearly reduced if $B < 1/2 B_{msy}$)

$MSY = 26.4$, 95% CL = 19.9 - 36.5; $B_{msy} = 308$, 95% CL = 187 - 498 (1000 tonnes)

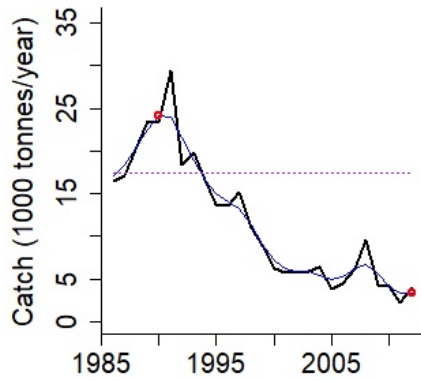
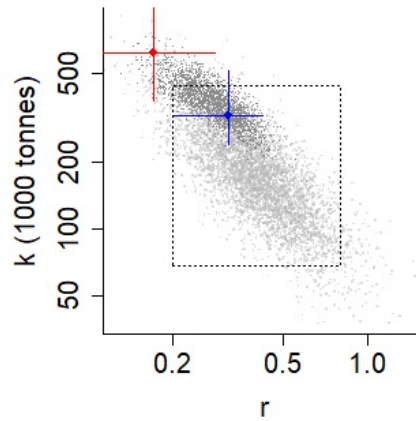
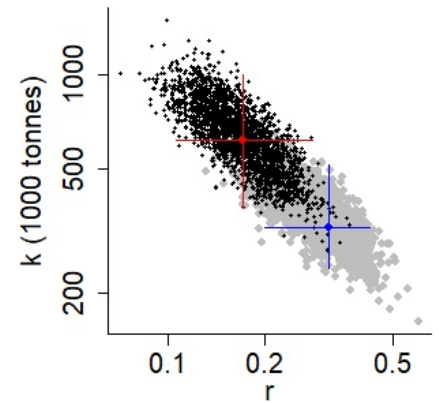
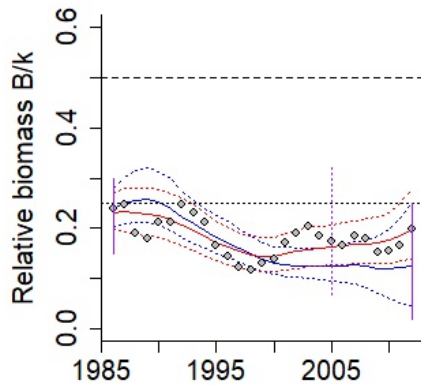
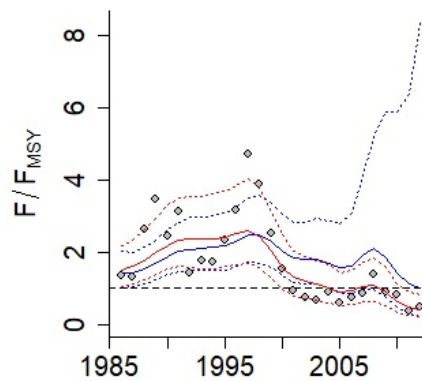
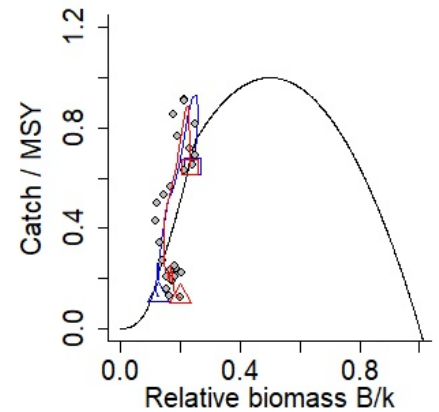
Biomass in last year = 122, 95% CL = 72.8 - 202 (1000 tonnes)

B/B_{msy} in last year = 0.397, 95% CL = 0.278 - 0.558

Fishing mortality in last year = 0.0274, 95% CL = 0.0154 - 0.0492

$F/F_{msy} = 0.401$, 95% CL = 0.212 - 0.825

Comment: NOAA Assessment model SS3. RF OK

A: Hogfish - Florida Keys - East Flor**B: Finding viable r-k****C: Analysis of viable r-k****D: Stock size****E: Exploitation rate****F: Equilibrium curve**

extext

Results of CMSY analysis conducted in JAGS

$r = 0.317$, 95% CL = 0.2 - 0.422; $k = 323$, 95% CL = 239 - 514 (1000 tonnes)

MSY = 25.6, 95% CL = 20 - 33.5 (1000 tonnes/year)

Relative biomass last year = 0.125 k , 95% CL = 0.0455 - 0.252

Exploitation $F/(r/2)$ in last year = 1.13

Results from Bayesian Schaefer model using catch and CPUE

$r = 0.172$, 95% CL = 0.106 - 0.283; $k = 616$, 95% CL = 373 - 995

r - k log correlation = -0.81

MSY = 26.4, 95% CL = 19.9 - 36.5 (1000 tonnes/year)

Relative biomass in last year = 0.125 k , 95% CL = 0.0455 - 0.252

Exploitation $F/(r/2)$ in last year = 0.598

$q = 3.27$, 95% CL = 2.06 - 5.23

Prior range of q = 1.13 - 28.9

Relative abundance data type = CPUE

Prior initial relative biomass = 0.15 - 0.3 expert

Prior intermediate relative biomass = 0.0678 - 0.323 in year 2005 default

Prior final relative biomass = 0.02 - 0.25 expert

Prior range for r = 0.2 - 0.8 default, prior range for k = 68.5 - 439 (1000 tonnes) default

Source for relative biomass:

NOAA Fisheries. (2020). Assessment Summary Data. Retrieved from www.st.nmfs.noaa.gov/stocksmart. 12/13/2020